

WINA BWANGU FINTECH SYSTEM - TEST PLAN AND EXECUTION

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1. Introduction

This test plan covers testing of Wina Bwangu prototype. The prototype is the first version maintained from Exercise 2 and extended with server-side implementation for Exercise 3. The purpose is to verify that the prototype meets functional requirements.

2. Test Objectives

- Validate client-side features: navigation, dropdown dependencies (Booth → Location → Services), TransactionID auto-generation, revenue calculation, form validation, toast notifications, responsive design.
- Validate server-side features: Express.js APIs, database layer, business logic validation.
- Ensure integration between client and server works correctly.

3. Test Scope

In Scope: Dashboard view, Transaction form, Booth Performance view, API endpoints (/api/booths, /api/services, /api/transactions, /api/dashboard, POST /api/transactions), responsive breakpoints (768px, 1024px).

4. Test Environment

OS: Windows 11 | Browser: Chrome 127 | IDE: VS Code 1.90 | Runtime: Node.js 20 LTS | Framework: Express.js 4.18.2 | Frontend: Tailwind CSS + Vanilla JS | Chart: Chart.js 4.4.0 via jsDelivr | Tools: DevTools Console, Postman, Manual testing, Node assertions

5. Unit Testing - Execution Results

Unit testing was performed for individual modules in isolation.

ID	Module	Input / Action	Expected	Actual	Status
UT-01	TransactionID	counter=1	WB0000001	WB0000001	PASS
UT-02	TransactionID	counter=171	WB0000171	WB0000171	PASS
UT-03	Revenue Airtel	964 * 0.05	48.2	48.2	PASS
UT-04	Revenue MTN	582 * 0.06	34.92	34.92	PASS

UT-05	Booth Filter	Select Wina4	3 services	3 services	PASS
UT-06	Booth Filter	Select Wina1	5 services	5 services	PASS
UT-07	Location	Select Wina3	Kabwata	Kabwata	PASS
UT-08	Rate Display	Select MTN Money	0.06	0.06	PASS
UT-09	Validation Empty	Submit empty form	Toast error	Toast shown	PASS
UT-10	Validation Amount	Amount = 0	Error toast	Error shown	PASS
UT-11	API POST	Wina1, Airtel, 500	201 Created	201 + ID	PASS
UT-12	API Invalid	Invalid booth Wina9	400 error	400 Invalid	PASS
UT-13	API GET Dash	GET /api/dashboard	JSON totals	Totals OK	PASS
UT-14	API GET Booths	GET /api/booths	6 booths	6 booths	PASS

Unit Testing Result: 14/14 PASSED (100%)

6. Regression Testing - Execution Results

Regression testing was performed after integrating backend and fixing Chart.js CDN.

ID	Feature	Change Made	Retest Action	Result	Status
RT-01	Chart.js	CDN to jsDelivr	Load dashboard	Charts render OK	PASS
RT-02	Navigation	Added backend	Click tabs	Tabs switch OK	PASS
RT-03	Dropdowns	database.js	Select booths	6 booths OK	PASS
RT-04	Responsive	Tailwind kept	Resize 768px	Stacks OK	PASS
RT-05	Toasts	Backend errors	Trigger toasts	Toasts OK	PASS
RT-06	Revenue	Client+Server	Compare calc	Match OK	PASS
RT-07	Table	Fetch API	Load table	Table OK	PASS
RT-08	Export CSV	Persistence	Click Export	CSV OK	PASS

RT-09	TxID	Backend counter	Create 2 tx	Increment OK	PASS
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Regression Testing Result: 9/9 PASSED (100%)

7. Summary

Type	Total	Passed	Failed	Pass Rate
Unit Testing	14	14	0	100%
Regression Testing	9	9	0	100%
TOTAL	23	23	0	100%

8. Conclusion

The prototype testing is completed successfully with 100% pass rate (23/23 tests passed). Unit testing confirms all modules work correctly. Regression testing confirms that after adding server-side prototype and fixing CDN, all client-side features still function without breakage. The prototype meets all requirements for Logbook Exercise 3.